



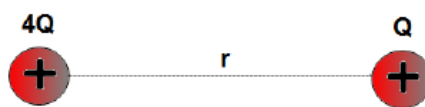
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Assignment (1)

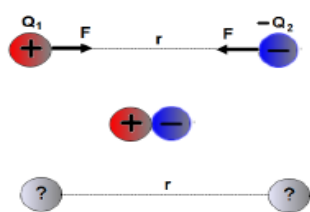
Electric force

No.	1	2	3	4	5	6	7	8	9	10
Answer										

Q1: Choose the correct answer

1. A plastic rod is rubbed with a piece of animal fur. The plastic rod acquires a negative charge during this process. Which of the following is true about the charge on the piece of fur?			
a. It acquires a positive charge but greater in magnitude than the rod	b. It acquires a positive charge but less in magnitude than the rod	c. It acquires a positive charge with the same magnitude as the rod	d. Does not acquire any charge
2. Two positive charges with magnitudes $4Q$ and Q are separated by a distance r. Which of the following statements is true?			
			
a. The charge with a greater magnitude exerts a larger force on the small charge	b. The charge with a greater magnitude exerts a smaller force on the small charge	c. The forces on each charge are the same in magnitude and opposite in direction	d. The forces on each charge are the same in magnitude and pointing in the same direction
3. Two positive charges Q_1 and Q_2 are separated by a distance r. The charges repel each other with a force F. If the magnitude of each charge is doubled and the distance stays unchanged what is the new force between the charges?			



a. F	b. $2F$	c. $4F$	d. $F/4$
4. Two positive charges Q_1 and Q_2 are separated by a distance r . The charges repel each other with a force F . If the distance between the charges is cut to one-fourth what is the new force acting on each charge?			
a. $16F$	b. $2F$	c. $4F$	d. $F/4$
5. Two charges Q_1 and $-Q_2$ are separated by a distance r . The charges attract each other with a force F . What is the new force between the charges if the distance is tripled?			
a. $3F$	b. $9F$	c. $F/6$	d. $F/9$
6. Two charges Q_1 and $-Q_2$ are separated by a distance r . The charges attract each other with a force F . What is the new force between the charges if the distance is cut to one-fourth and the magnitude of each charge is doubled?			
a. $16F$	b. $64F$	c. $48F$	d. $F/16$
7. A positively charged sphere with a charge of $+8Q$ is separated from a negatively charged sphere $-2Q$ by a distance r . The spheres briefly touch each other and move to the original distance r . What is the new charge on each sphere after they move to the distance r ?			
			
a. Q	b. $2Q$	c. $3Q$	d. $Q/4$
8. SI unit of permittivity of free space is			
a. $C^2 N^{-2} m^{-1}$	b. $C^2 N^{-1} m^{-2}$	c. $C^2 N m^2$	d. $C N^{-1} m^{-2}$
9. Where an electric charge experiences a force, there is an			
a. Electric Field	b. Magnetic Field	c. Electric current	d. Conventional current
10. The force between two charges is 120 N . If the distance between the charges is doubled, the force will be			
a. 60 N	b. 30 N	c. 40 N	d. 15 N

Best wishes

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